Value-Informed Schedule Trimming for Accelerated Sequential Monte Carlo Guidance in Discrete Diffusion

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Abstract

Reward-guided discrete diffusion enables test-time alignment with downstream objectives, but full Sequential Monte Carlo (SMC) guidance is computationally expensive. While unguided steps only require the standard generative model transition, active guidance steps additionally pays for reward evaluations, and gradient computations in some models. Sparse guidance reduces this cost by guiding only a subset of timesteps, yet existing schedules are largely hand-designed, such as early, late, or uniform guidance. This is fundamentally suboptimal because the guidance timing is task-dependent: some objectives need early global steering, some need late local correction, and some need both. We propose VISTA-SMC, a training-free pre-inference scheduler that automatically selects guided steps under a fixed budget. The core idea is to view sparse guidance as approximating the ideal full-guidance probability path. Although the ground-truth full-guidance marginals are intractable, we derive an estimable upper bound on their average ℓ_1 path deviation and minimize the resulting surrogate by dynamic programming using a few warmup full-guidance runs. The warmup cost is amortized across subsequent generations, so after schedule selection VISTA-SMC has the same per-generation complexity as any sparse method with the same budget. Across text, biological sequence, and image generation tasks, VISTA-SMC discovers task-specific schedules, strongly outperforms hand-designed sparse schedules at the same budget, closely approaches or matches full guidance on primary metrics, and reaches up to $3.44\times$ faster inference in the main experiments.

1 Introduction

Discrete diffusion models [2] have emerged as a powerful framework for generating structured objects such as text, biological sequences, and discrete images. In these domains, practitioners often prioritize not only the likelihood under a pretrained model but also alignment with external objectives, such as safety in language, biological activity in DNA, or constraint satisfaction. Consequently, a primary goal is to sample from a reward-tilted distribution that steers the pretrained model toward higher expected rewards while preserving its generative capabilities.

Several reward-tilted guidance techniques for discrete domains have recently been introduced [23, 29]. However, these independent sampling methods—which generate samples along separate trajectories—often rely on derivative-based approximations that can introduce significant error and bias [17].

To address the limitation, Sequential Monte Carlo (SMC) methods have been proposed. SMC begins by generating multiple candidate particles using a proposal policy. To ensure alignment with the reward-tilted distribution, it employs a resampling mechanism: particles with high value functions are duplicated, while those with low values are discarded (see Figure 1-(b)). This framework provides

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theoretical guarantees that particles will follow the exact reward-tilted distribution [17, 26, 22, 10, 15, 20].

Despite their theoretical elegance, the computational cost of SMC guidance remains a bottleneck in practical applications. Each guided denoising step requires not only a pretrained diffusion transition but also an evaluation by an external reward model; moreover, gradient-based proposals often necessitate additional backward passes. Applying this correction at every timestep t can dominate the total wall-clock time. Consequently, sparse guidance—where only a subset of $T' < T$ timesteps is guided—is essential for efficient inference-time alignment, where T denotes the total number of diffusion steps This approach can reduce the total reward and gradient computation to a fraction of T'/T (see Table 2 for a detailed breakdown).

However, a central question remains under-explored: *which* steps should be guided? Existing sparse schedules are typically heuristic-based and fixed: guiding only at the beginning, at the end, within a middle interval, or uniformly. Such rigid rules are often brittle. For instance, a visual task may require early global steering to avoid suboptimal modes, whereas a text generation task might benefit from a mixture of early anchors and late-stage corrections. Similarly, biological design may necessitate both coarse early control and fine-grained functional refinement. Ultimately, a schedule that is optimal for one reward landscape can be suboptimal, or even detrimental, to another.

To mitigate this issue, we propose VISTA-SMC (Value-Informed Schedule Trimming for Accelerated Sequential Monte Carlo), a training-free, plug-and-play procedure for identifying an optimal budgeted schedule of guided steps. By minimizing the discrepancy between the full-guidance trajectory and the reduced-guidance path, VISTA-SMC characterizes the intrinsic properties of a given task and identifies a highly effective schedule tailored to its specific reward landscape.

Specifically, VISTA-SMC addresses sparse SMC guidance as a full-path approximation problem: it chooses the guided-step set (i.e., $\mathcal{G}$) so that the reduced-guidance path stays close to the ideal full-guidance path (i.e., $\{\pi_t^*\}_{t=T}^0$). Directly optimizing this gap is impossible because the ground-truth full-guidance marginals are not available as explicit distributions over the discrete state space. Our main observation is that the gap admits a tractable upper bound expressed through value expectations along the full-guidance path. These expectations can be estimated from a small number of warmup full-guidance runs; the resulting budgeted objective is then solved exactly by dynamic programming. The selected schedule is reused for all later generations with the same model, reward, budget, and proposal family, so the warmup cost is amortized and the per-generation comparison to sparse heuristics is fair.

Our contributions are ordered by conceptual importance:

1. **Automatic guidance timing.** We introduce a training-free scheduler that chooses when SMC guidance is needed under a fixed budget, replacing early/late/middle/uniform rules.
2. **Full-guidance gap formulation.** We view sparse SMC guidance as approximating the ideal full-guidance probability path, not only the final marginal. This is crucial: final-only correction may be asymptotically valid yet can collapse diversity with finite particles.
3. **Tractable gap minimization without ground-truth marginals.** To our knowledge, VISTA-SMC is the first sparse-SMC scheduler to target the gap to full-guidance marginals without accessing those intractable marginals. We prove an estimable upper bound on average ℓ_1 path deviation and minimize it under a budget.
4. **Amortized and fair inference-time comparison.** Schedule construction uses a few warmup full-guidance runs. After this one-time cost, VISTA-SMC uses the same guided-step count and per-generation forward complexity as heuristic sparse schedules at the same budget.
5. **Task-specific empirical schedules.** On Binary MNIST, toxic text, and DNA sequence generation, VISTA-SMC finds reward-dependent schedules, outperforms hand-designed sparse baselines at the same budget, and yields large speedups over full guidance, including $3.44\times$ in the main toxic-text SMC-grad experiment.

2 Related Work

Discrete and Masked Diffusion. Diffusion models define a forward process that transforms a data distribution into a tractable prior and learn reverse transitions for generation. While continuous

diffusion models typically use Gaussian perturbations and score-based reverse dynamics [13, 31], discrete diffusion models define Markov transitions over categorical or masked states [2, 28]. Masked diffusion language models instantiate this idea by using a special [MASK] token as the high-noise state and iteratively unmasking or denoising tokens. Our work focuses on inference-time reward guidance for such reverse processes, with a particular emphasis on addressing the challenges of discrete state spaces.

Sequential Monte Carlo Methods. SMC samplers propagate particles through a proposal policy and use resampling to correct bias, effectively aligning the particles with the target reward-tilted distribution [7, 22, 6]. Recent work has adapted SMC to diffusion inference for tasks such as continuous and discrete diffusion steering, test-time alignment, inverse problems, and tree-structured search [26, 30, 12, 1, 36, 14, 11, 5]. While these methods primarily focus on *how* particles are weighted or proposed, VISTA-SMC is complementary: it automatically identifies the most critical timesteps for guidance, determining *when* these expensive corrections should be executed to maximize overall efficiency.

Inference-time Acceleration of Diffusion Guidance. Guidance acceleration for diffusion models often focuses on reducing the frequency or computational overhead of classifier-free guidance (CFG) evaluations through fixed intervals, adaptive rules, caching, or multirate integration [4, 21, 18, 16]. While these methods demonstrate that continuous guidance is not always necessary, they are primarily designed for continuous-image CFG or network-evaluation reuse. In contrast, VISTA-SMC is specifically formulated for reward-guided SMC in discrete state spaces and provides a theoretical bound on the probability-path deviation. Table 1 summarizes these key distinctions.

Table 1: Comparison of VISTA-SMC with existing guidance acceleration methods. Discrete: Designed for discrete state spaces; Reward: Handles external reward functions; Theoretical Guarantee: Provides a rigorous theoretical justification for the acceleration schedule.

Method	Discrete	Reward	Theoretical Guarantee
Adaptive Guidance [4]	×	×	×
FasterCache [21]	×	×	×
Tortoise-and-Hare [18]	×	×	✓
Guidance Interval [16]	×	×	×
VISTA-SMC (Ours)	✓	✓	✓

3 Preliminaries

3.1 Reference Reverse Process

Let $\mathcal{X}$ be a discrete state space. A pretrained reverse diffusion model defines a reference path:

$$p_T(x_T) \prod_{t=0}^{T-1} p_{t|t+1}(x_t|x_{t+1}), \tag{1}$$

with one-step reverse kernels $p_{t|t+1}$ and marginals p_t. For $0 \leq s \leq t \leq T$, we denote by $p_{s|t}$ the composed transition from x_t to x_s. In a masked diffusion model, x_T is typically a fully masked sequence, and x_0 is a clean sequence.

3.2 Reward-Tilted Full-Guidance Path

Given a reward function $r(x_0)$ for a sample x_0, our objective is to maximize the expected reward while regularizing the divergence from the pretrained model's prior. Specifically, we aim to find an optimal policy $\{\pi_t\}_{t=T}^{0}$ that maximizes:

$$\mathbb{E}_{\{\pi_t\}_{t=T}^{0}} \left[r(x_0) - \alpha \sum_{t=0}^{T-1} \mathrm{KL}(\pi_{t|t+1}(\cdot|x_{t+1}) \parallel p_{t|t+1}(\cdot|x_{t+1})) - \alpha \mathrm{KL}(\pi_T(\cdot) \parallel p_T(\cdot)) \right], \tag{2}$$

where $\alpha > 0$ is a temperature parameter controlling the trade-off between reward maximization and prior preservation [9, 32, 33]. The closed-form solution to this optimization problem is:

$$\pi^*_{t|t+1}(x_t|x_{t+1}) = \frac{V_t(x_t)}{V_{t+1}(x_{t+1})} p_{t|t+1}(x_t|x_{t+1}), \quad t \in \{0, \dots, T-1\} \tag{3}$$

$$\pi^*_t(x_t) = \frac{1}{Z} V_t(x_t) p_t(x_t), \tag{4}$$

where $Z := \mathbb{E}_{x_0 \sim p_0}[R(x_0)]$ is the partition function. Here, $R(x_0) := \exp(r(x_0)/\alpha)$ represents the exponentiated reward, and $V_t(x_t)$ is the exponentiated value function defined as:

$$V_t(x_t) := \mathbb{E}_{x_0 \sim p_{0|t}(\cdot|x_t)} \left[R(x_0) \right]. \tag{5}$$

Although the value function $v_t(x_t) := \alpha \log V_t(x_t)$ is standard in the literature [33, 34], the exponentiated notation V_t simplifies our derivation.

For a detailed derivation, we refer the reader to [33] or our Appendix A.2.

Definition 3.1. We call π^*_t the **full-guidance path**. It is the gold-standard path that VISTA-SMC aims to approximate.

3.3 SMC Proposal Families

Algorithm 1 SMC guidance

1: $x_T^{(i)} \sim \pi^*_T(x_T) \quad (i = 1, 2, \cdots, N)$
2: **for** $t = T-1, T-2, \cdots, 0$ **do**
3: Propagation: $x_t^{(i)} \overset{\text{iid}}{\sim} q_{t|t+1}(x_t|x_{t+1}^{(i)}) \quad (i = 1, 2, \cdots, N)$
4: Resampling: Duplicate particles with high V_t and discard those with low V_t to ensure $x_t^{(i)} \sim \pi^*_t(x_t)$
5: **end for**

SMC guidance propagates particles through a proposal policy $q_{t|t+1}$ and utilizes the resampling to align the particle distribution with the target path π^*_t (Algorithm 1; see Algorithm 4 for details). Generally, the closer $q_{t|t+1}$ is to $\pi^*_{t|t+1}$, the higher the sampling efficiency. We evaluate VISTA-SMC on three proposal families established in recent discrete SMC literature [26]:

- **SMC-base** uses the pretrained transition: $q_{t|t+1}(x_t|x_{t+1}) = p_{t|t+1}(x_t|x_{t+1})$.
- **SMC-grad** uses a first-order Taylor approximation of the value function to steer the proposal, yielding $q_{t|t+1}(x_t|x_{t+1}) \propto p_{t|t+1}(x_t|x_{t+1}) \exp(\frac{1}{\alpha} \nabla v_{t+1}(x_{t+1}) \cdot x_t)$.
- **SMC-amot** uses an amortized proposal network $q_{t|t+1}$ trained to approximate $\pi^*_{t|t+1}$. We assume this proposal satisfies Assumption A.6.

VISTA-SMC is proposal-agnostic; it does not modify the underlying proposal family. Instead, it strategically selects which timesteps should incur the computational overhead of value estimation and resampling to maximize efficiency.

4 VISTA-SMC: Value-Informed Schedule Trimming

In this section, we introduce Reduced SMC Guidance, a framework designed to accelerate the inference of guided diffusion models by strategically skipping guidance steps.

4.1 Sparse guidance as full-path approximation

Let $\mathcal{T} = \{0, 1, \cdots, T-1\}$ denote the set of reverse timesteps. A sparse schedule is a subset $\mathcal{G} \subset \mathcal{T}$ with budget $|\mathcal{G}| = T'$. Under this framework, reward guidance is applied if and only if $t \in \mathcal{G}$, as detailed in Algorithm 2.

Algorithm 2 Diffusion Guidance Reduction

$x_T^{(i)} \sim \pi_T^*(x_T) \quad (i = 1, 2, \cdots, N)$
for $t = T-1, T-2, \cdots, 0$ **do**
  Guidance is ON $\leftarrow t \in \mathcal{G}$
  if Guidance is ON **then**
    Propagation: $x_t^{(i)} \overset{\text{iid}}{\sim} q_{t|t+1}(x_t|x_{t+1}^{(i)}) \quad (i = 1, 2, \cdots, N)$
    Resampling: Adjust particles $\{x_t^{(i)}\}$ to ensure $x_t^{(i)} \sim \pi_t^*(x_t)$
  else
    Propagation: $x_t^{(i)} \overset{\text{iid}}{\sim} p_{t|t+1}(x_t|x_{t+1}^{(i)}) \quad (i = 1, 2, \cdots, N)$
  end if
end for

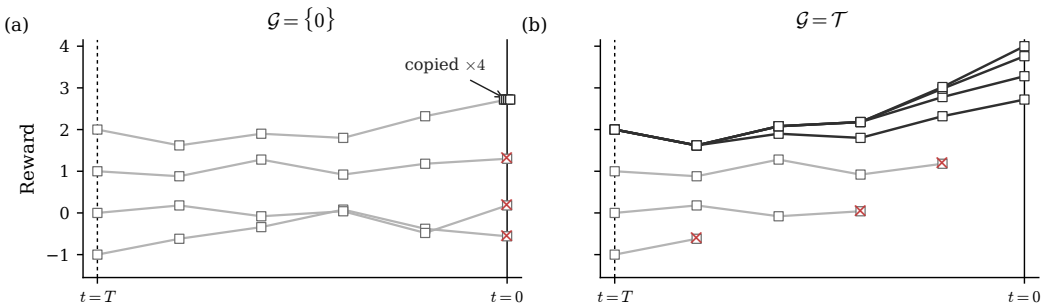

Figure 1: Impact of guidance schedules on particle distributions. A minimal schedule (e.g., $\mathcal{G} = \{0\}$) is asymptotically correct as $N \to \infty$, but with finite particles, it leads to diversity collapse through the duplication of a few terminal particles. In contrast, full guidance ($\mathcal{G} = \mathcal{T}$) maintains intermediate marginals close to the reward-tilted path, ensuring both high reward and sample diversity.

For any $t \in \mathcal{T}$, we define the most recent active step $\lceil t \rceil$ as:

$$\lceil t \rceil := \min\{s \in \mathcal{G} \cup \{T\} \mid s \geq t\}. \tag{6}$$

If $t \in \mathcal{G}$, then $\lceil t \rceil = t$ and the sampler performs a guided SMC correction. If $t \notin \mathcal{G}$, it propagates from $\lceil t \rceil$ to t using unguided reference transitions. This observation leads to the following definition:

Definition 4.1. For a sparse schedule $\mathcal{G}$, the **reduced-guidance path** is

$$\pi_t^{\mathcal{G}}(x_t) := \sum_{x_{\lceil t \rceil} \in \mathcal{X}} \pi_{\lceil t \rceil}^*(x_{\lceil t \rceil}) p_{t|\lceil t \rceil}(x_t|x_{\lceil t \rceil}). \tag{7}$$

A naive objective would be to find $\mathcal{G}$ such that only the final marginal $\pi_0^{\mathcal{G}}$ matches the target π_0^*. Theoretically, if the number of particles N is infinite, even a minimal schedule like $\mathcal{G} = \{0\}$ is sufficient to recover the exact reward-tilted distribution at the final step. However, this ignores a critical practical challenge: in the finite-N regime, delaying guidance until the end causes the sampler to postpone all reward information. This leads to diversity collapse and poor reward (see Fig. 1). To mitigate this degradation in practical sampling, we instead aim to minimize the deviation between the full-guidance and reduced-guidance paths across all timesteps.

Definition 4.2. The **Total Deviation** for a schedule $\mathcal{G}$ is defined as the average ℓ_1 distance between the full-guidance and reduced-guidance paths:

$$\mathbf{TD}(\mathcal{G}) := \frac{1}{T} \sum_{t=0}^{T-1} \|\pi_t^* - \pi_t^{\mathcal{G}}\|_1 = \frac{1}{T} \sum_{t=0}^{T-1} \sum_{x_t \in \mathcal{X}} |\pi_t^*(x_t) - \pi_t^{\mathcal{G}}(x_t)|. \tag{8}$$

This definition formalizes the core objective of VISTA-SMC: schedule selection is transformed from a heuristic choice of intervals into a principled, budgeted optimization to minimize the gap between the sparse sampling path and the ideal full-guidance probability path.

4.2 A tractable upper bound

A major obstacle in directly optimizing $\mathbf{TD}(\mathcal{G})$ is the intractability of the full-guidance path π_t^*. The following theorem provides a computable surrogate based on value expectations that can be estimated from a small number of warmup runs.

Theorem 4.3. *Assume $0 \leq R(x_0) \leq R_{\max} < \infty$. For any schedule $\mathcal{G} \subset \mathcal{T}$,*

$$\mathbf{TD}(\mathcal{G}) \leq \mathbf{UTD}(\mathcal{G}) := \frac{2}{T} \sum_{t \in \mathcal{T} \setminus \mathcal{G}} [1 - \frac{1}{R_{max}} \mathbb{E}_{x_{\lceil t \rceil} \sim \pi^*_{\lceil t \rceil}} [V_{\lceil t \rceil}(x_{\lceil t \rceil})]]. \tag{9}$$

The proof of this inequality is provided in Section A.3.

Our goal is to identify a schedule $\mathcal{G}$ that minimizes $\mathbf{UTD}$. This approach—optimizing a tractable upper bound when the primary objective is intractable—is a well-established strategy in machine learning [3, 25, 24].

For a fixed budget of guidance steps $T' = |\mathcal{G}|$, minimizing $\mathbf{UTD}$ is equivalent to maximizing the following objective:

$$S(\mathcal{G}) := \sum_{t \in \mathcal{T} \setminus \mathcal{G}} \mathbb{E}_{x_{\lceil t \rceil} \sim \pi^*_{\lceil t \rceil}} \left[V_{\lceil t \rceil}(x_{\lceil t \rceil}) \right]. \tag{10}$$

Intuitively, if the state at the most recent active step $x_{\lceil t \rceil}$ already exhibits a high expected reward, subsequent unguided propagation is less likely to deviate significantly from the optimal path.

Crucially, this optimization does not require knowledge of the full marginals $\pi_t^*(x_t)$, nor the constant $R_{\max}$; it relies solely on the Monte Carlo estimation of $\mathbb{E}_{x_t \sim \pi_t^*} [V_t(x_t)]$.

4.3 Description of the Method

VISTA-SMC estimates $\mathbb{E}_{x_t \sim \pi_t^*} [V_t(x_t)]$ by running M full-guidance warmup simulations. Let $\hat{V}_t^{(i,m)}$ be the value estimate of i-th particle at timestep t during the m-th warmup run. We approximate the mean value as:

$$\mathbb{E}_{x_t \sim \pi_t^*} [V_t(x_t)] \approx \hat{V}_t := \frac{1}{MN} \sum_{m=1}^{M} \sum_{i=1}^{N} \hat{V}_t^{(i,m)}. \tag{11}$$

The empirical schedule $\hat{\mathcal{G}}$ is then obtained by solving:

$$\hat{\mathcal{G}} \in \arg \max_{\mathcal{G} \subset \mathcal{T} : |\mathcal{G}| = T', 0 \in \mathcal{G}} \hat{S}(\mathcal{G}), \quad \hat{S}(\mathcal{G}) := \sum_{t \in \mathcal{T} \setminus \mathcal{G}} \hat{V}_{\lceil t \rceil} \tag{12}$$

(The constraint $0 \in \mathcal{G}$ ensures $\pi_0^{\mathcal{G}} = \pi_0^*$.)

The optimization in Eq. (12) is solved exactly by dynamic programming in $O(T^2 T')$ time (see Algorithm 6), which is negligible for the horizons used here. Once $\hat{\mathcal{G}}$ is selected, VISTA-SMC utilizes this sparse schedule for all subsequent generations to accelerate inference.

4.4 Computational Time Analysis

Table 2 summarizes the computational cost per generation after the schedule $\mathcal{G}$ has been determined. For VISTA-SMC applied to SMC-amot, we modify line 8 of Algorithm 2 to $x_t^{(i)} \sim q_{t|t+1}(x_t | x_{t+1}^{(i)})$. This is motivated by the fact that the computational costs of p and q are virtually identical, yet q remains a more effective proposal for guided sampling. A formal justification ensuring that this modification preserves the deviation bound in Theorem 4.3 is provided in Theorem A.7.

While the M warmup simulations represent an initial overhead not required by heuristic methods, this cost is rapidly amortized as the number of generated samples increases. In large-scale deployment, the efficiency gains from the optimized sparse far outweigh the one-time cost of value estimation. (See A.6 for details.)

Table 2: Per-generation cost after schedule selection. Here, p is a pretrained reverse-model evaluation, q is an amortized proposal, r is a reward-model evaluation, ∇v is a value-gradient evaluation, N is the number of particles, T is the diffusion horizon, T' is the guidance budget, and J is the number of rollout samples for value estimation (see Algorithm 5 for details).

Variant	**Full Guidance**	**Sparse Guidance**
SMC-BASE	$NT \times p,\ NTJ \times r$	$NT \times p,\ NT'J \times r$
SMC-GRAD	$NT \times p,\ NTJ \times r,\ NT \times \nabla v$	$NT \times p,\ NT'J \times r,\ NT' \times \nabla v$
SMC-AMOT	$NT \times q,\ NTJ \times r$	$NT \times q,\ NT'J \times r$

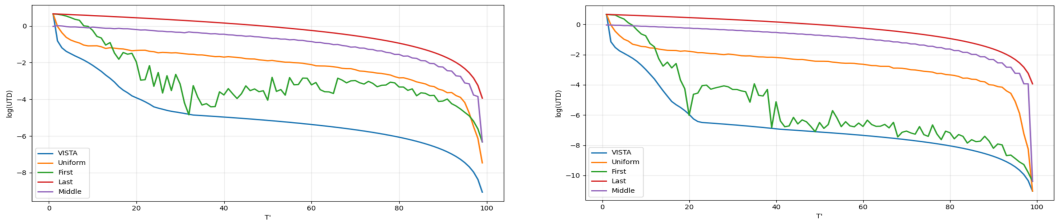

Figure 2: **UTD** vs. Budget (T'). The y-axis represents $\log(\mathbf{UTD})$ and the x-axis denotes the budget T'. Among all compared methods, VISTA-SMC demonstrates the most rapid convergence of $\mathbf{UTD}$ toward zero (i.e., $\log(\mathbf{UTD}) \to -\infty$), effectively minimizing the path discrepancy even under highly constrained step budgets. (a) SMC-base and (b) SMC-grad settings are shown respectively.

5 Experiments

To evaluate the effectiveness of VISTA-SMC, we focus our experiments on Masked Diffusion Models (MDMs), the dominant architecture for the discrete domains studied in this work. (Results for other discrete diffusion models are provided in Appendix B.2.) Our implementation is built upon the codebase of [26]. We compare VISTA-SMC against full guidance and four heuristic sparse schedule baselines: **First, Last, Middle, and Uniform**. Unless otherwise stated, all reported times correspond to the per-generation cost after schedule selection; warmup costs are reported separately. Comprehensive experimental details are available in Appendix B.

5.1 Synthetic Experiment: Binary MNIST

We first evaluate VISTA-SMC on Binary MNIST using the MDM framework. The reward is defined as $r(x_0) = \log p(c|x_0)$, where c represents the condition "the digit is 4". We use a total horizon of $T = 100$ with $\alpha = 1.0$, $N = 20$ particles, a guidance budget of $T' = 25$, $J = 1$ rollout for value estimation, and $M = 3$ warmup simulations. A sample x_0 is considered "valid" if $r(x_0) > -0.1$. To measure diversity, "Unique Valid" counts the number of valid samples, where samples with a Jaccard similarity ≥ 0.75 are treated as identical. Results are averaged over 100 independent trials. The preparation phase (warmup and dynamic programming) took 9.83s for SMC-base and 21.6s for SMC-grad. For evaluation, the $\mathbf{UTD}$ was computed with $R_{\text{max}} = 1$.

As shown in Table 3, for SMC-grad, VISTA-SMC reduces the runtime from 4.74s to 2.41s (a 1.97× speedup) while maintaining nearly the same number of unique valid samples. In contrast, Late and Middle schedules suffer from diversity collapse, demonstrating that heuristic reduction of guidance steps is insufficient for maintaining sample quality. The schedule (see Figure 6) is early-heavy, which is consistent with the need to steer the global digit identity before the process becomes irreversible.

5.2 Toxic Text Generation

Toxic text generation is a high-dimensional discrete task where the reward model must guide the sampler against a strong language prior. The reward is defined as $r(x_0) = \log p(c|x_0)$ for a target toxicity condition. Without guidance, the base model generates toxic content with only a 0.8% success rate. We evaluate performance using four metrics: (i) Toxic, measured by the inference-time reward model [19]; (ii) Toxic-ext, measured by an external classifier [8]; (iii) Perplexity, evaluated via

Table 3: Binary MNIST results.

Variant	Schedule	Time [s] ↓	Unique valid ↑	Reward ↑	UTD ↓
SMC-BASE	Full	1.76 ± 0.11	16.78 ± 4.12	−0.0447 ± 0.2735	–
	First	1.71 ± 0.14	17.12 ± 3.32	−0.1456 ± 0.3656	7.882×10^{-2}
	Last	1.69 ± 0.13	1.88 ± 1.16	−0.6996 ± 1.9815	1.462
	Middle	1.70 ± 0.13	2.43 ± 1.96	−0.6063 ± 1.8203	7.524×10^{-1}
	Uniform	1.68 ± 0.12	14.41 ± 3.80	−0.0535 ± 0.1092	2.309×10^{-1}
	VISTA-SMC	**1.69 ± 0.12**	**16.90 ± 3.35**	**−0.0636 ± 0.2191**	$\mathbf{1.132 \times 10^{-2}}$
SMC-GRAD	Full	4.74 ± 0.26	17.00 ± 2.94	−0.0021 ± 0.0042	–
	First	2.40 ± 0.06	17.60 ± 2.82	−0.0663 ± 0.1887	1.366×10^{-2}
	Last	2.36 ± 0.06	1.92 ± 1.14	−0.6968 ± 2.1626	1.462
	Middle	2.37 ± 0.06	2.28 ± 1.70	−0.3522 ± 1.3745	7.228×10^{-1}
	Uniform	2.42 ± 0.09	14.60 ± 3.68	−0.0127 ± 0.0263	1.448×10^{-1}
	VISTA-SMC	**2.41 ± 0.07**	**17.11 ± 3.02**	**−0.0124 ± 0.0296**	$\mathbf{1.448 \times 10^{-3}}$

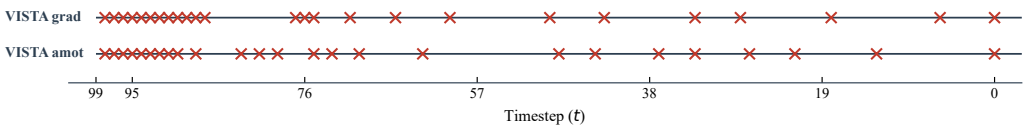

Figure 3: Picture of $\mathcal{G}$ in toxic text

GPT2-XL [27]; and (iv) Distinct n-gram ratios. We set $T = 99, \alpha = 0.2, N = 8, T' = 25, J = 4$, and $M = 3$. The preparation phase required 634.3s for SMC-grad and 114.9s for SMC-amot.

As shown in Table 4, VISTA-SMC significantly accelerates inference while maintaining strong reward alignment. For SMC-grad, VISTA-SMC achieves a 3.44× speedup (from 209.91s to 60.99s) while retaining 100% toxicity. Similarly, for SMC-amot, it provides a 1.91× speedup (from 36.95s to 19.35s) and yields higher toxicity than all heuristic baselines. Figure 3 illustrates that the selected schedule exhibits a hybrid structure of 'First' and 'Uniform' patterns, rather than a single contiguous interval, suggesting that VISTA-SMC identifies non-trivial guidance requirements across the sampling trajectory.

5.3 Biological Sequence Design

DNA sequence design evaluates whether the scheduler can balance functional reward with biological plausibility. We adopt the generation and reward models from [35] and follow their evaluation protocol using five metrics: (i) Likelihood, (ii) Predicted Activity, (iii) ATAC-Acc, (iv) 3-mer Pearson, and (v) Jaspar Spearman. We set $T = 127, \alpha = 0.2, N = 8, T' = 64, J = 4$, and $M = 3$.

Table 4: Toxic text generation.

Variant	Schedule	Time [s] ↓	Toxic [%] ↑	Toxic-ext [%] ↑	Dist-1/2/3 [%] ↑	PPL ↓	$\hat{S}(\mathcal{G})$ ↑
SMC-GRAD	Full	209.91 ± 0.85	100.0	90.8	51.3/79.7/85.6	117.6	–
	First	61.06 ± 0.55	73.3	74.2	60.0/93.1/95.9	103.7	2.30×10^{-7}
	Last	61.10 ± 0.53	86.7	57.5	20.3/32.5/39.8	132.8	7.34×10^{-13}
	Middle	61.10 ± 0.54	95.8	81.7	49.5/80.4/88.4	103.4	1.82
	Uniform	60.13 ± 0.55	94.2	85.8	53.3/81.5/86.5	100.8	9.62
	VISTA-SMC	**60.99 ± 0.66**	**100.0**	**100.0**	**80.8/94.8/94.8**	**119.7**	**11.04**
SMC-AMOT	Full	36.95 ± 0.58	100.0	87.5	27.0/43.5/49.4	58.6	–
	First	19.25 ± 0.36	35.8	60.8	57.1/88.7/93.6	68.4	2.61×10^{-8}
	Last	19.24 ± 0.21	46.7	42.5	13.7/19.0/21.4	89.3	7.34×10^{-13}
	Middle	19.25 ± 0.24	73.3	76.7	44.6/75.2/84.6	78.3	7.72
	Uniform	19.23 ± 0.27	85.8	92.5	30.1/48.0/54.0	83.7	18.0
	VISTA-SMC	**19.35 ± 0.32**	**87.5**	**95.0**	**29.6/48.0/55.5**	**67.4**	**20.8**

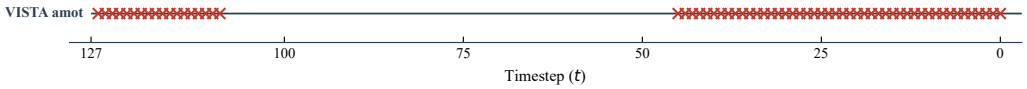

Figure 4: Picture of $\mathcal{G}$ in DNA generation

Table 5: DNA sequence generation

Schedule	Time [s] ↓	Likelihood ↑	Pred-Activity ↑	ATAC-Acc ↑	3-mer Pearson ↑	JASPAR Spearman ↑	$\hat{S}(\mathcal{G})$ ↑
Full	11.28 ± 0.04	-253.469 ± 10.703	5.509 ± 0.959	0.858 ± 0.288	0.771 ± 0.111	0.630 ± 0.165	–
First	6.20 ± 0.03	-257.613 ± 8.780	2.035 ± 0.947	0.292 ± 0.240	0.784 ± 0.162	0.790 ± 0.077	7.13×10^{12}
Last	6.19 ± 0.03	-258.191 ± 9.473	5.308 ± 1.090	0.808 ± 0.341	0.706 ± 0.113	0.602 ± 0.121	6.30×10^{1}
Middle	6.19 ± 0.03	-259.471 ± 8.471	3.427 ± 1.439	0.467 ± 0.364	0.737 ± 0.122	0.721 ± 0.079	3.01×10^{14}
Uniform	6.21 ± 0.02	-250.889 ± 30.090	5.441 ± 1.232	0.842 ± 0.334	0.762 ± 0.127	0.602 ± 0.114	1.85×10^{16}
VISTA-SMC	**6.19 ± 0.03**	$\mathbf{-255.076 \pm 10.725}$	$\mathbf{5.601 \pm 0.782}$	$\mathbf{0.858 \pm 0.266}$	$\mathbf{0.751 \pm 0.096}$	$\mathbf{0.622 \pm 0.122}$	$\mathbf{7.32 \times 10^{17}}$

Results are averaged over 15 independent seeds, yielding 120 generated samples per schedule. The preparation phase required 34.9s.

As shown in Table 5, VISTA-SMC reduces the runtime from 11.28s to 6.19s (a 1.82× speedup) while matching the ATAC accuracy of full guidance. Notably, it even improves the predicted activity from 5.509 to 5.601 in this finite-particle setting. These two metrics are the primary indicators of reward alignment. The optimized schedule, visualized in Figure 4, exhibits a hybrid of 'First' and 'Last' patterns. This is consistent with the biological requirements of the task: early steps establish the broad sequence architecture, while terminal steps refine functional motifs and cell-type specificity

5.4 Discussion

The optimized schedules identified by VISTA-SMC are distinctly task-specific, reflecting the unique dynamics of each reward landscape. For instance, the early-heavy pattern in MNIST suggests that global modes are determined in the initial stages, whereas the bimodal structure in DNA design indicates a "coarse-to-fine" requirement—establishing structural validity early and functional refinement late. These non-intuitive patterns are difficult to capture with manual heuristics, which often lead to the catastrophic failures observed in fixed schedules.

While the UTD bound is conservative, it consistently identifies robust schedules that avoid the pitfalls of sub-optimal timing. Our findings demonstrate that efficient alignment is fundamentally a problem of strategic timing rather than just guidance strength. By treating the schedule as a data-driven component, VISTA-SMC provides a principled framework for navigating the trade-off between alignment quality and computational efficiency.

6 Limitation

Despite its effectiveness, VISTA-SMC has two primary limitations. First, it incurs an initial warmup overhead (e.g., $M = 3$ runs) to estimate the value function. Although this cost is amortized over large-scale generations, it may be less efficient for single-shot sampling tasks. Second, the UTD bound is conservative. As an upper bound, it may lead to sub-optimal schedules in scenarios with highly non-linear reward landscapes or where finite-particle diversity is the primary bottleneck. Future work could focus on tighter bounds or adaptive, real-time schedule adjustments to further refine the efficiency-quality trade-off.

7 Conclusion

We introduced VISTA-SMC, a training-free scheduler for sparse reward-guided SMC in discrete diffusion. By minimizing an estimable upper bound on path deviation, VISTA-SMC identifies task-specific guidance schedules that maximize alignment quality under a fixed computational budget. Our experiments across image, text, and biological domains demonstrate that automatic schedule selection can match or even slightly exceed full-guidance performance while substantially reducing inference latency. Ultimately, this work shows that for efficient discrete diffusion alignment, choosing *when* to guide is as important as designing *how* to guide.

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    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: Our research does not involve crowdsourcing nor experiments with human subjects.

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    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

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- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

A Theoretical Details

A.1 Background

Lemma A.1.

$$V_t(x_t) = \mathbb{E}_{x_s \sim p_{s|t}(\cdot|x_t)}[V_s(x_s)] \tag{13}$$

where $0 \leq s < t \leq T$. *Also,*

$$\mathbb{E}_{x_t \sim p_t}[V_t(x_t)] = \mathbb{E}_{x_0 \sim p_0}[R(x_0)]. \tag{14}$$

Proof.

$$V_t(x_t) = \mathbb{E}_{x_0 \sim p_{0|t}(\cdot|x_t)}[R(x_0)] \tag{15}$$

comes from the definition of value function.

$$\mathbb{E}_{x_s \sim p_{s|t}(\cdot|x_t)}[V_s(x_s)] = \mathbb{E}_{x_s \sim p_{s|t}(\cdot|x_t)}[\mathbb{E}_{x_0 \sim p_{0|s}(\cdot|x_s)}[R(x_0)]] \tag{16}$$

$$= \mathbb{E}_{x_0 \sim p_{0|t}(\cdot|x_t)}[R(x_0)] = V_t(x_t). \tag{17}$$

$$\mathbb{E}_{x_T \sim p_T}[V_T(x_T)] = \mathbb{E}_{x_T \sim p_T}[\mathbb{E}_{x_0 \sim p_{0|T}(\cdot|x_T)}[R(x_0)]] = \mathbb{E}_{x_0 \sim p_0}[R(x_0)]. \tag{18}$$

□

Theorem A.2. $\{\pi_t\}_{t=T}^0$ *that maximizes*

$$\mathbb{E}_{\{\pi_t\}_{t=T}^0}[r(x_0) - \alpha \sum_{t=T-1}^{0} \mathrm{KL}(\pi_{t|t+1}(\cdot|x_{t+1})||p_{t|t+1}(\cdot|x_{t+1})) - \alpha \mathrm{KL}(\pi_T(\cdot)||p_T(\cdot))] \tag{19}$$

($\alpha > 0$*) is*

$$\pi_t(x_t) = \frac{1}{Z} V_t(x_t) p_t(x_t). \quad (Z := \mathbb{E}_{x_0 \sim p_0}[R(x_0)]) \tag{20}$$

Proof. We define the Markov process $\{\pi_t^*\}_{t=T}^0$ as

$$\pi_T^*(x_T) = \frac{1}{Z} V_T(x_T) p_T(x_T), \tag{21}$$

$$\pi_{t|t+1}^*(x_t|x_{t+1}) = \frac{V_t(x_t) p_{t|t+1}(x_t|x_{t+1})}{V_{t+1}(x_{t+1})} \quad (t = 0, 1, \cdots, T-1). \tag{22}$$

(These distributions are well-defined(normalized) by Lemma A.1.)

We observe

$$\mathbb{E}_{\{\pi_t\}_{t=T}^0}[r(x_0) - \alpha \sum_{t=T-1}^{0} \mathrm{KL}(\pi_{t|t+1}(\cdot|x_{t+1})||p_{t|t+1}(\cdot|x_{t+1})) - \alpha \mathrm{KL}(\pi_T(\cdot)||p_T(\cdot))] \tag{23}$$

$$= \mathbb{E}_{x_0 \sim \pi_0}[r(x_0)] - \alpha \sum_{t=T-1}^{0} \mathbb{E}_{x_t, x_{t+1} \sim \pi_{t,t+1}}[\log \frac{\pi_{t|t+1}(x_t|x_{t+1})}{p_{t|t+1}(x_t|x_{t+1})}] - \alpha \mathbb{E}_{x_T \sim \pi_T}[\log \frac{\pi_T(x_T)}{p_T(x_T)}] \tag{24}$$

$$= \mathbb{E}_{x_T \sim \pi_T} \mathbb{E}_{x_{T-1} \sim \pi_{T-1|T}(\cdot|x_T)} \cdots \mathbb{E}_{x_0 \sim \pi_{0|1}(\cdot|x_1)}$$
$$[r(x_0) - \alpha \sum_{t=T-1}^{0} \log \frac{\pi_{t|t+1}(x_t|x_{t+1})}{p_{t|t+1}(x_t|x_{t+1})} - \alpha \log \frac{\pi_T(x_T)}{p_T(x_T)}]. \tag{25}$$

If we use the recurrence relation

$$\mathbb{E}_{x_s \sim \pi_{s|s+1}(\cdot|x_{s+1})}[v_s(x_s) - \alpha \log \frac{\pi_{s|s+1}(x_s|x_{s+1})}{p_{s|s+1}(x_s|x_{s+1})}] \tag{26}$$

$$= \mathbb{E}_{x_s \sim \pi_{s|s+1}(\cdot|x_{s+1})}[v_{s+1}(x_{s+1}) - \alpha \log \frac{\pi_{s|s+1}(x_s|x_{s+1})}{\pi_{s|s+1}^*(x_s|x_{s+1})}] \tag{27}$$

$$= v_{s+1}(x_{s+1}) - \alpha \mathrm{KL}(\pi_{s|s+1}(\cdot|x_{s+1})||\pi_{s|s+1}^*(\cdot|x_{s+1})) \tag{28}$$

consequently, Eq. (25) could be simplified into

$$\mathbb{E}_{x_T\sim\pi_T}[v_T(x_T) - \alpha \log \frac{\pi_T(x_T)}{p_T(x_T)}] - \alpha \sum_{t=T-1}^{0} \mathbb{E}_{x_{t+1}\sim\pi_{t+1}}[\mathrm{KL}(\pi_{t|t+1}(\cdot|x_{t+1})||\pi^*_{t|t+1}(\cdot|x_{t+1}))] \tag{29}$$

$$=\alpha \log Z - \alpha \sum_{t=T-1}^{0} \mathbb{E}_{x_{t+1}\sim\pi_{t+1}}[\mathrm{KL}(\pi_{t|t+1}(\cdot|x_{t+1})||\pi^*_{t|t+1}(\cdot|x_{t+1}))] - \alpha \mathrm{KL}(\pi_T(\cdot)||\pi^*_T(\cdot)). \tag{30}$$

We now easily see that this objective is minimized iff $\pi_T = \pi^*_T$ and $\pi_{t|t+1} = \pi^*_{t|t+1}$ for all $t = 0, 1, \cdots, T-1$.

The final step is proving

$$\pi^*_t(x_t) = \frac{1}{Z} V_t(x_t) p_t(x_t). \tag{31}$$

This is satisfied at $t = T$. If

$$\pi^*_{t+1}(x_{t+1}) = \frac{1}{Z} V_{t+1}(x_{t+1}) p_t(x_{t+1}), \tag{32}$$

then

$$\pi^*_t(x_t) = \sum_{x_{t+1}} \pi^*_{t+1}(x_{t+1}) \pi^*_{t|t+1}(x_t|x_{t+1}) \tag{33}$$

$$= \sum_{x_{t+1}} \frac{V_{t+1}(x_{t+1}) p_{t+1}(x_{t+1})}{Z} \frac{V_t(x_t) p_{t|t+1}(x_t|x_{t+1})}{V_{t+1}(x_{t+1})} \tag{34}$$

$$= \frac{1}{Z} V_t(x_t) p_t(x_t). \tag{35}$$

Proof ends by induction.

□

Algorithm 3 Multinomial resampling

1: **Input:** $\{x^{(i)}\}_{i=1}^N, \{w^{(i)}\}_{i=1}^N$
2: **for** $k \leftarrow 1, 2, \cdots, N$ **do**
3: $\quad i_k \overset{\text{iid}}{\sim} \mathrm{Cat}(i : \frac{w^{(i)}}{\sum_{n=1}^N w^{(n)}})$
4: $\quad y^{(k)} \leftarrow x^{(i_k)}$
5: **end for**
6: **Output:** $\{y^{(k)}\}_{k=1}^N, \{1\}_{k=1}^N$

A.2 Reduced SMC algorithm

Algorithm 3 defines the idealized multinomial resampling operator utilized in our analysis. In practice, more efficient variants such as systematic, adaptive, or partial resampling may be employed; we refer the reader to [22] for a comprehensive review of these methods.

Lemma A.3 (Importance Sampling)**.** *Let $(x^{(i)}, \tilde{x}^{(i)}) \overset{iid}{\sim} \rho(x, \tilde{x})$, and let $w(x, \tilde{x}) \geq 0$ be a weight function such that $w^{(i)} = w(x^{(i)}, \tilde{x}^{(i)})$. As $N \to \infty$, the output particles $\{y^{(k)}\}_{k=1}^N$ of Algorithm 3 satisfy:*

$$y^{(k)} \overset{iid}{\sim} \pi(x) \propto \sum_{\tilde{x}} w(x, \tilde{x}) p(x, \tilde{x}). \tag{36}$$

Proof. For any test function $\phi(x)$,

$$\mathbb{E}_{x^{(i)},\tilde{x}^{(i)}\overset{\text{iid}}{\sim}\rho(x,\tilde{x})}[\phi(y^{(k)})] = \mathbb{E}_{x^{(i)},\tilde{x}^{(i)}\overset{\text{iid}}{\sim}\rho(x,\tilde{x})}[\frac{\frac{1}{N}\sum_{i=1}^N w(x^{(i)}, \tilde{x}^{(i)})\phi(x^{(i)})}{\frac{1}{N}\sum_{n=1}^N w(x^{(n)}, \tilde{x}^{(n)})}]. \tag{37}$$

Algorithm 4 Full SMC guidance

1: **for** $i \leftarrow 1, 2, \cdots, N$ **do**
2: $\quad x_T^{(i)} \overset{\text{iid}}{\sim} q_T(x_T)$
3: $\quad x_0^{(i,j)} \overset{\text{iid}}{\sim} p_{0|T}(x_0|x_T^{(i)}) \quad (j = 1, 2, \cdots, J)$
4: $\quad \hat{V}_T^{(i)} \leftarrow \frac{1}{J}\sum_{j=1}^{J} R(x_0^{(i,j)})$
5: $\quad w_T^{(i)} \leftarrow p_T(x_T^{(i)}) \times \hat{V}_T^{(i)} / q_T(x_T^{(i)})$
6: **end for**
7: $\{x_T^{(i)}\}_{i=1}^N, \{w_T^{(i)}\}_{i=1}^N \leftarrow \text{Resample}(\{x_T^{(i)}\}_{i=1}^N, \{w_T^{(i)}\}_{i=1}^N)$
8: **for** $t \leftarrow T-1, T-2, \cdots, 0$ **do**
9: $\quad$ **if** $t \in \mathcal{G}$ **then**
10: $\quad\quad$ **for** $i \leftarrow 1, 2, \cdots, N$ **do**
11: $\quad\quad\quad x_t^{(i)} \overset{\text{iid}}{\sim} q_{t|t+1}(x_t|x_{t+1}^{(i)})$
12: $\quad\quad\quad x_0^{(i,j)} \overset{\text{iid}}{\sim} p_{0|t}(x_0|x_t^{(i)}) \quad (j = 1, 2, \cdots, J)$
13: $\quad\quad\quad \hat{V}_t^{(i)} \leftarrow \frac{1}{J}\sum_{j=1}^{J} R(x_0^{(i,j)})$
14: $\quad\quad\quad w_t^{(i)} \leftarrow \frac{p_{t|t+1}(x_t^{(i)}|x_{t+1}^{(i)})}{q_{t|t+1}(x_t^{(i)}|x_{t+1}^{(i)})} \times \frac{\hat{V}_t^{(i)}}{\hat{V}_{\lceil t+1 \rceil}^{(i)}} \times w_{\lceil t+1 \rceil}^{(i)}$
15: $\quad\quad$ **end for**
16: $\quad\quad \{x_t^{(i)}\}_{i=1}^N, \{w_t^{(i)}\}_{i=1}^N \leftarrow \text{Resample}(\{x_t^{(i)}\}_{i=1}^N, \{w_t^{(i)}\}_{i=1}^N)$
17: $\quad$ **else**
18: $\quad\quad$ **for** $i \leftarrow 1, 2, \cdots, N$ **do**
19: $\quad\quad\quad x_t^{(i)} \overset{\text{iid}}{\sim} p_{t|t+1}(x_t|x_{t+1}^{(i)})$
20: $\quad\quad$ **end for**
21: $\quad$ **end if**
22: **end for**

By the law of large numbers, as $N \to \infty$,

$$\frac{1}{N}\sum_{n=1}^{N} w(x^{(n)}, \tilde{x}^{(n)}) \xrightarrow{\text{a.s.}} \mathbb{E}_{x,\tilde{x}\sim\rho}[w(x,\tilde{x})] \tag{38}$$

$$\frac{1}{N}\sum_{i=1}^{N} w(x^{(i)}, \tilde{x}^{(i)})\phi(x^{(i)}) \xrightarrow{\text{a.s.}} \mathbb{E}_{x,\tilde{x}\sim\rho}[w(x,\tilde{x})\phi(x)] = \mathbb{E}_{x,\tilde{x}\sim\rho}[w(x,\tilde{x})] \times \mathbb{E}_{x\sim\pi}[\phi(x)]. \tag{39}$$

Thus, we get

$$\mathbb{E}_{x^{(i)},\tilde{x}^{(i)} \overset{\text{iid}}{\sim} \rho(x,\tilde{x})}[\phi(y^{(k)})] \overset{N\to\infty}{\to} \mathbb{E}_{x\sim\pi}[\phi(x)], \tag{40}$$

so $y^{(k)} \sim \pi$.

Lastly, regarding the independence of $\{y^{(k)}\}_{k=1}^N$, the multinomial resampling process draws each $y^{(k)}$ independently given the particle set. As $N \to \infty$, this process is equivalent to i.i.d. sampling from the target distribution π, as the empirical distribution converges to the true π.

□

Theorem A.4. *Say $N, J \to \infty$. At every guided time $t \in \mathcal{G} \cup \{T\}$, the particles in Algorithm 5 are distributed according to the ideal guided target distribution, i.e., $x_t^{(i)} \overset{iid}{\sim} \pi_t^* \quad (i = 1, 2, \cdots, N)$.*

Proof. Since $J \to \infty$, the Monte Carlo value estimate $\hat{V}_t^{(i)}$ converges almost surely to the true value function $V_t(x_t^{(i)})$ by the strong law of large numbers.

Base Case: At $t = T$, the particles are sampled from q_T and weighted by $w_T \propto p_T(x_T)V_T(x_T)/q_T(x_T)$. By Lemma A.3, after resampling at Line 7, we have $x_T^{(i)} \overset{\text{iid}}{\sim} \pi_T^*(x_T) \propto p_T(x_T)V_T(x_T)$.

Algorithm 5 Sparse SMC guidance

1: **for** $i \leftarrow 1, 2, \cdots, N$ **do**
2: $\quad x_T^{(i)} \overset{\text{iid}}{\sim} q_T(x_T)$
3: $\quad x_0^{(i,j)} \overset{\text{iid}}{\sim} p_{0|T}(x_0|x_T^{(i)}) \quad (j = 1, 2, \cdots, J)$
4: $\quad \hat{V}_T^{(i)} \leftarrow \frac{1}{J}\sum_{j=1}^{J} R(x_0^{(i,j)})$
5: $\quad w_T^{(i)} \leftarrow p_T(x_T^{(i)}) \times \hat{V}_T^{(i)} / q_T(x_T^{(i)})$
6: **end for**
7: $\{x_T^{(i)}\}_{i=1}^N, \{w_T^{(i)}\}_{i=1}^N \leftarrow \text{Resample}(\{x_T^{(i)}\}_{i=1}^N, \{w_T^{(i)}\}_{i=1}^N)$
8: **for** $t \leftarrow T-1, T-2, \cdots, 0$ **do**
9: $\quad$ **if** $t \in \mathcal{G}$ **then**
10: $\quad\quad$ **for** $i \leftarrow 1, 2, \cdots, N$ **do**
11: $\quad\quad\quad x_t^{(i)} \overset{\text{iid}}{\sim} q_{t|t+1}(x_t|x_{t+1}^{(i)})$
12: $\quad\quad\quad x_0^{(i,j)} \overset{\text{iid}}{\sim} p_{0|t}(x_0|x_t^{(i)}) \quad (j = 1, 2, \cdots, J)$
13: $\quad\quad\quad \hat{V}_t^{(i)} \leftarrow \frac{1}{J}\sum_{j=1}^{J} R(x_0^{(i,j)})$
14: $\quad\quad\quad w_t^{(i)} \leftarrow \frac{p_{t|t+1}(x_t^{(i)}|x_{t+1}^{(i)})}{q_{t|t+1}(x_t^{(i)}|x_{t+1}^{(i)})} \times \frac{\hat{V}_t^{(i)}}{\hat{V}_{\lceil t+1 \rceil}^{(i)}} \times w_{\lceil t+1 \rceil}^{(i)}$
15: $\quad\quad$ **end for**
16: $\quad\quad \{x_t^{(i)}\}_{i=1}^N, \{w_t^{(i)}\}_{i=1}^N \leftarrow \text{Resample}(\{x_t^{(i)}\}_{i=1}^N, \{w_t^{(i)}\}_{i=1}^N)$
17: $\quad$ **else**
18: $\quad\quad$ **for** $i \leftarrow 1, 2, \cdots, N$ **do**
19: $\quad\quad\quad x_t^{(i)} \overset{\text{iid}}{\sim} p_{t|t+1}(x_t|x_{t+1}^{(i)})$
20: $\quad\quad$ **end for**
21: $\quad$ **end if**
22: **end for**

Inductive Step: Assume that $x_{\lceil t+1 \rceil}^{(i)} \overset{\text{iid}}{\sim} \pi_{\lceil t+1 \rceil}^*(x_{\lceil t+1 \rceil})$.

Then, just before the line 16,

$$(x_t^{(i)}, x_{t+1}^{(i)}, x_{\lceil t+1 \rceil}^{(i)}) \overset{\text{iid}}{\sim} q_{t|t+1}(x_t|x_{t+1}) \cdot p_{t+1|\lceil t+1 \rceil}(x_{t+1}|x_{\lceil t+1 \rceil}) \cdot \frac{p_{\lceil t+1 \rceil}(x_{\lceil t+1 \rceil}) V_{\lceil t+1 \rceil}(x_{\lceil t+1 \rceil})}{Z}. \tag{41}$$

Just after the line 16,

$$\begin{aligned}(x_t^{(i)}, x_{t+1}^{(i)}, x_{\lceil t+1 \rceil}^{(i)}) &\overset{\text{iid}}{\sim} q_{t|t+1}(x_t|x_{t+1}) p_{t+1|\lceil t+1 \rceil}(x_{t+1}|x_{\lceil t+1 \rceil}) \frac{p_{\lceil t+1 \rceil}(x_{\lceil t+1 \rceil}) V_{\lceil t+1 \rceil}(x_{\lceil t+1 \rceil})}{Z} \\ &\quad \times \frac{p_{t|t+1}(x_t|x_{t+1})}{q_{t|t+1}(x_t|x_{t+1})} \frac{V_t(x_t)}{V_{\lceil t+1 \rceil}(x_{\lceil t+1 \rceil})} \qquad (42) \\ &= \frac{1}{Z} p_{t,t+1,\lceil t+1 \rceil}(x_t, x_{t+1}, x_{\lceil t+1 \rceil}) V_t(x_t). \qquad (43)\end{aligned}$$

By Lemma A.3, we may conclude that

$$x_t^{(i)} \overset{\text{iid}}{\sim} \pi_t^*(x_t). \tag{44}$$

The proof follows by induction for all $t \in \mathcal{G}$. □

A.3 Probability path distance

We first record a simple inequality.

Lemma A.5. *Random variable $X \in \mathbb{R}$ satisfies $0 \le X \le M$ and $\mu = \mathbb{E}[X]$. Then,*

$$\mathbb{E}[|X - \mu|] \le 2\mu \left(1 - \frac{\mu}{M}\right). \tag{45}$$

Proof. Since

$$0 = \mathbb{E}[X - \mu] = \mathbb{E}[(X - \mu)\mathbf{1}(X > \mu) + (X - \mu)\mathbf{1}(X \leq \mu)],$$

we get the following identity:

$$\mathbb{E}[(X - \mu)\mathbf{1}(X > \mu)] = \mathbb{E}[(\mu - X)\mathbf{1}(X \leq \mu)]. \tag{46}$$

Thus,

$$\begin{aligned}\mathbb{E}[|X - \mu|] &= \mathbb{E}[(X - \mu)\mathbf{1}(X > \mu)] \\ &\quad + \mathbb{E}[(\mu - X)\mathbf{1}(X \leq \mu)] \\ &= 2\mathbb{E}[(\mu - X)\mathbf{1}(X \leq \mu)] \\ &\leq 2\mathbb{E}[\mu(1 - X/M)] = 2\mu(1 - \mu/M).\end{aligned}$$

Applying the expectation to both sides:

$$\begin{aligned}\mathbb{E}[|X - \mu|] &\leq 2\mathbb{E}\left[\mu\left(1 - \frac{X}{M}\right)\right] \\ &= 2\mu\left(1 - \frac{\mathbb{E}[X]}{M}\right) = 2\mu\left(1 - \frac{\mu}{M}\right).\end{aligned}$$

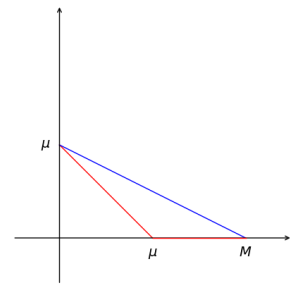

Figure 5: Illustration of $(\mu - X)\mathbf{1}(X \leq \mu) \leq \mu(1 - \frac{X}{M})$

□

Proof of Theorem 4.3. If $t \in \mathcal{G}$, then $\pi_t^* = \pi_t^{\mathcal{G}}$, so

$$\mathbf{TD}(\mathcal{G}) = \frac{1}{T}\sum_{t \in \mathcal{T} - \mathcal{G}} \|\pi_t^* - \pi_t^{\mathcal{G}}\|_1. \tag{47}$$

By Lemma A.1,

$$V_{\lceil t \rceil}(x_{\lceil t \rceil}) = \mathbb{E}_{x_t \sim p_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil})}[V_t(x_t)]. \tag{48}$$

Thus,

$$\|\pi_t^* - \pi_t^{\mathcal{G}}\|_1 = \sum_{x_t \in \mathcal{X}} |\pi_t^*(x_t) - \sum_{x_{\lceil t \rceil} \in \mathcal{X}} \pi_{\lceil t \rceil}^*(x_{\lceil t \rceil}) p_{t|\lceil t \rceil}(x_t|x_{\lceil t \rceil})| \tag{49}$$

$$= \frac{1}{Z}\sum_{x_t \in \mathcal{X}} |\sum_{x_{\lceil t \rceil} \in \mathcal{X}} p_{\lceil t \rceil}(x_{\lceil t \rceil}) p_{t|\lceil t \rceil}(x_t|x_{\lceil t \rceil})(V_t(x_t) - V_{\lceil t \rceil}(x_{\lceil t \rceil}))| \tag{50}$$

$$\leq \frac{1}{Z}\mathbb{E}_{x_{\lceil t \rceil} \sim p_{\lceil t \rceil}, x_t \sim p_{t|\lceil t \rceil}(\cdot|x_t)}|V_t(x_t) - V_{\lceil t \rceil}(x_{\lceil t \rceil})|. \tag{51}$$

Applying Lemma A.5 with $X = V_t(x_t)$ and $\mu = V_{\lceil t \rceil}(x_{\lceil t \rceil})$ gives

$$\mathbb{E}_{x_t \sim p_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil})}[|V_t(x_t) - V_{\lceil t \rceil}(x_{\lceil t \rceil})|] \leq 2V_{\lceil t \rceil}(x_{\lceil t \rceil})(1 - \frac{V_{\lceil t \rceil}(x_{\lceil t \rceil})}{R_{\max}}). \tag{52}$$

Therefore,

$$\|\pi_t^* - \pi_t^{\mathcal{G}}\|_1 \leq \frac{2}{Z}\mathbb{E}_{x_{\lceil t \rceil} \sim p_{\lceil t \rceil}}[V_{\lceil t \rceil}(x_{\lceil t \rceil})(1 - \frac{V_{\lceil t \rceil}(x_{\lceil t \rceil})}{R_{\max}})] \tag{53}$$

$$= 2(1 - \frac{1}{R_{\max}}\mathbb{E}_{x_{\lceil t \rceil} \sim \pi_{\lceil t \rceil}^*}[V_{\lceil t \rceil}(x_{\lceil t \rceil})]). \tag{54}$$

□

A.4 Amortized proposals during skipped steps

The derivation of the main UTD bound (Theorem 4.3) assumes that skipped steps employ the reference transition p. However, in our SMC-amot experiments, skipped intervals may instead be propagated using an amortized proposal q, as discussed in Section 4.4. We show that the following condition is sufficient to maintain the same UTD bound under this setting.

Assumption A.6. For all $0 \leq t < s \leq T$,

$$\mathbb{E}_{x_s \sim \pi_s^*} \|q_{t|s}(\cdot|x_s) - \pi^*_{t|s}(\cdot|x_s)\|_1 \leq \mathbb{E}_{x_s \sim \pi_s^*} \|p_{t|s}(\cdot|x_s) - \pi^*_{t|s}(\cdot|x_s)\|_1. \tag{55}$$

Intuitively, Assumption A.6 implies that the amortized proposal $q_{t|s}$ provides a closer approximation to the optimal posterior $\pi^*_{t|s}$ than the unconditional transition $p_{t|s}$. This assumption is empirically grounded, as $q_{t|s}$ is explicitly trained to minimize the divergence from the guided distribution.

Theorem A.7. *Under the assumption A.6,*

$$\frac{1}{T}\sum_{t=0}^{T-1} \|\pi^*_t(\cdot) - \sum_{x_{\lceil t \rceil} \in \mathcal{X}} \pi^*_{\lceil t \rceil}(x_{\lceil t \rceil}) q_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil})\|_1 \leq \mathbf{UTD}(\mathcal{G}) \tag{56}$$

holds.

Proof.

$$\begin{aligned}
&\|\pi^*_t(x_t) - \sum_{x_{\lceil t \rceil}} \pi^*_{\lceil t \rceil}(x_{\lceil t \rceil}) q_{t|\lceil t \rceil}(x_t|x_{\lceil t \rceil})\|_1 && (57)\\
&\leq \mathbb{E}_{x_{\lceil t \rceil} \sim \pi^*_{\lceil t \rceil}} \|\pi^*_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil}) - q_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil})\|_1 && (58)\\
&\leq \mathbb{E}_{x_{\lceil t \rceil} \sim \pi^*_{\lceil t \rceil}} \|\pi^*_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil}) - p_{t|\lceil t \rceil}(\cdot|x_{\lceil t \rceil})\|_1 \quad \text{by A.6} && (59)\\
&= \frac{1}{Z}\mathbb{E}_{x_{\lceil t \rceil} \sim p_{\lceil t \rceil}, x_t \sim p_{t|\lceil t \rceil}(\cdot|x_t)} |V_t(x_t) - V_{\lceil t \rceil}(x_{\lceil t \rceil})|, && (60)
\end{aligned}$$

which matches Eq. (51). Summing over t yields the desired bound. □

A.5 DP algorithm

We aim to find $\mathcal{G}$ with $0 \in \mathcal{G}$ (to ensure $x_0 \sim \pi^*_0$) that maximizes

$$\hat{\mathbf{S}}(\mathcal{G}) := \sum_{t \in \mathcal{T} \setminus \mathcal{G}} \hat{V}_{\lceil t \rceil} \tag{61}$$

where $T' = |\mathcal{G}|$ is a given constant.

Define

$$dp[s][i] := \max_{\mathcal{G}'} \sum_{t \in \mathcal{T} \setminus \mathcal{G}'; t > s} \hat{V}_{\lceil t \rceil} \tag{62}$$

where

$$\mathcal{G}' = \{s = t_i < \cdots < t_1 < T\}. \tag{63}$$

We can observe the recurrence relation

$$\begin{aligned}
dp[s][1] &= (T - 1 - s)\hat{V}_T && (64)\\
dp[s][i] &= \max_{s < s' \leq T-i+1} [dp[s'][i-1] + (s' - s - 1)\hat{V}_{s'}]. \quad (i \geq 2) && (65)
\end{aligned}$$

Thus, Algorithm A.5 can compute the optimal $\mathcal{G}$.

Algorithm 6 DP algorithm

```
Initialize: dp[s][i] ← 0 for all s, i
for s ← T' − 1, T', ··· , T − 1 do
    dp[s][1] ← (T − 1 − s) V̂_T
end for
for i ← 2, 3, ··· , T' do
    for s ← T' − i, ··· , T − i do
        for s' ← s + 1 to T − i + 1 do
            c ← dp[s'][i − 1] + (s' − s − 1) V̂_{s'}
            if c > dp[s][i] then
                dp[s][i] ← c
                parent[s][i] ← s'
            end if
        end for
    end for
end for
t_{T'} ← 0
for j ← T' − 1, T' − 2, ··· , 1 do
    t_j ← parent[t_{j+1}][j + 1]
end for
return G = {t_1, t_2, ··· , t_{T'}}
```

A.6 Amortization of Warmup Overhead

To quantify the overhead, let Δt_{full} denote the time required for a single generation (of N samples) under full guidance, and Δt_{sparse} the time under a sparse schedule with budget T'. Suppose we aim to generate a total of K batches (i.e., KN samples). A heuristic sparse method, which requires no warmup, incurs a total time of:

$$T_H(K) = K\Delta t_{\text{sparse}}. \tag{66}$$

In contrast, VISTA-SMC first performs M warmup runs before switching to the optimized sparse schedule. Its total time for $K > M$ is:

$$T_V(K) = M\Delta t_{\text{full}} + (K - M)\Delta t_{\text{sparse}}. \tag{67}$$

While $T_V(K) > T_H(K)$ for any finite K, the relative overhead vanishes asymptotically:

$$\lim_{K\to\infty} \frac{T_H(K)}{T_V(K)} = \lim_{K\to\infty} \frac{K\Delta t_{\text{sparse}}}{M\Delta t_{\text{full}} + (K - M)\Delta t_{\text{sparse}}} = 1. \tag{68}$$

Thus, for a sufficiently large number of samples, the average cost per sample of VISTA-SMC is identical to that of a heuristic sparse method, while providing a more principled and effective guidance schedule.

B Experimental Details

B.1 Experimental Setup

All experiments were conducted on a single NVIDIA A6000 GPU. We emphasize that, unlike some SMC-based methods that sample N candidates and select only the one with the highest reward, we retained all generated samples to evaluate the overall distribution and alignment quality.

For Binary MNIST, we employed the masked diffusion model (MDM) from [28]. For Toxic Text Generation, we used the pretrained MDLM [28] alongside the reward model from [19]. For Biological Sequence Design, we utilized the pretrained MDM and reward model provided by [35]. The specific training hyperparameters and implementation details follow [26]. The resampling configurations for each domain are summarized in Table 6.

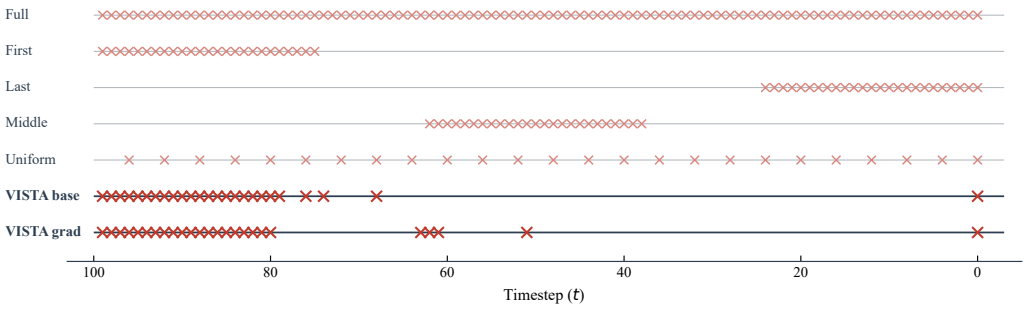

Figure 6: Picture of $\mathcal{G}$ in BinaryMNIST

Table 6: Resampling configurations for each experiment. Adaptive resampling is triggered when the Effective Sample Size (ESS) falls below the specified threshold. Partial resampling indicates the number of particles replaced during a resampling step.

Experiment	Adaptive ESS threshold	Partial resampling	Systematic resampling
Binary MNIST	0.95	$\lfloor \frac{N}{2} \rfloor$	yes
Toxic Text	0.98	N	yes
Biology Design	0.98	$\lfloor \frac{N}{2} \rfloor$	yes

B.2 Generalization to Other Discrete Diffusion Models

To demonstrate the versatility of VISTA-SMC beyond masked diffusion, we evaluate its performance on ReMDM and Uniform diffusion models using the Binary MNIST task. For ReMDM, we employ a max-capped remasking schedule with $\eta_{\text{cap}} = 0.1$. Both models are guided via the SMC-grad method, maintaining the same experimental configuration as described in Section 5.1. Results are averaged over 20 independent runs.

As summarized in Table 7, VISTA-SMC achieves significant speedups of 2.09× (ReMDM) and 1.95× (Uniform) while consistently outperforming or matching hand-designed heuristics. Specifically, in ReMDM, both 'First' and VISTA-SMC —which exhibit the lowest UTD—yield the best alignment. Similarly, in Uniform diffusion, the 'Uniform' and 'VISTA-SMC' schedules achieve superior performance (attaining the highest Unique Valid and Reward, respectively), again corresponding to their minimal UTD values. These results demonstrate that the UTD metric serves as a reliable surrogate for predicting actual task performance across different discrete diffusion architectures.

B.3 Sensitivity to Warmup Sample Size

To verify the reliability of our value function estimation in Eq. (11), we conduct a sensitivity study on the number of warmup simulations $M = 1, 2, \cdots, 10$. The goal is to demonstrate that a minimal number of runs (specifically $M = 3$) is sufficient to identify a stable and effective guidance schedule $\mathcal{G}$. As illustrated in Figures 8–12, the optimized schedules exhibit consistency across various M values. While larger M can provide further refinement, the marginal changes in the selected timesteps are negligible compared to the increased computational cost of the preparation phase. Therefore, we

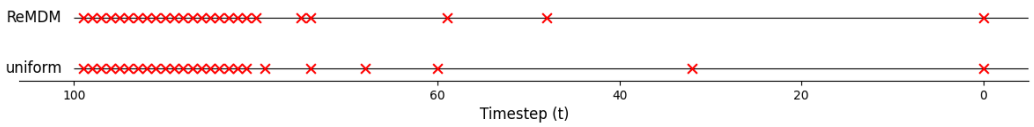

Figure 7: Picture of $\mathcal{G}$ in Table 7

Table 7: Binary MNIST ReMDM, uniform results.

Variant	Schedule	Time [s] ↓	Unique valid ↑	Reward ↑	UTD ↓
ReMDM	Full	5.34 ± 0.60	18.95 ± 1.50	-0.0005 ± 0.0008	–
	First	2.53 ± 0.19	18.25 ± 1.45	-0.5213 ± 0.4222	6.17×10^{-3}
	Last	2.52 ± 0.18	6.55 ± 3.82	-0.0103 ± 0.0266	1.46
	Middle	2.56 ± 0.22	18.05 ± 2.24	-0.0416 ± 0.0808	7.23×10^{-1}
	Uniform	2.60 ± 0.22	17.75 ± 3.65	-0.0031 ± 0.0048	1.44×10^{-1}
	VISTA-SMC	$\mathbf{2.56 \pm 0.18}$	$\mathbf{18.70 \pm 1.45}$	$\mathbf{-0.0131 \pm 0.0244}$	1.44×10^{-3}
Uniform	Full	4.88 ± 0.17	17.95 ± 2.35	-0.0236 ± 0.0325	–
	First	2.41 ± 0.12	10.85 ± 4.80	-4.8749 ± 2.9848	8.16×10^{-1}
	Last	2.38 ± 0.11	6.15 ± 4.73	-0.6448 ± 2.1527	1.45
	Middle	2.38 ± 0.05	17.10 ± 3.09	-0.2646 ± 1.0023	7.31×10^{-1}
	Uniform	2.55 ± 0.10	17.25 ± 3.26	-0.0905 ± 0.2154	4.50×10^{-1}
	VISTA-SMC	$\mathbf{2.50 \pm 0.12}$	$\mathbf{16.95 \pm 2.61}$	$\mathbf{-0.0277 \pm 0.0746}$	1.76×10^{-1}

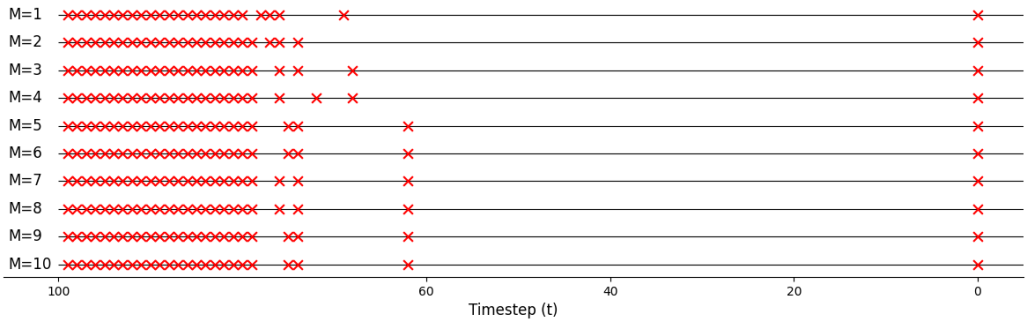

Figure 8: Optimized guidance schedules $\mathcal{G}$ for varying M (Binary MNIST, SMC-base)

adopt $M = 3$ as a standard for all experiments to ensure a highly efficient, low-overhead transition to VISTA-SMC.

B.4 Sensitivity to Guidance Budget T'

To demonstrate that our findings hold across various computational budgets, we evaluate VISTA-SMC under different guidance constraints, $T' \in \{10, 35\}$, complementing the $T' = 25$ results presented in the main text.

For the Binary MNIST task, we employ the same MDM architecture and hyperparameters as described in Section 5.1, with results averaged over 20 independent runs. As summarized in Table 8,

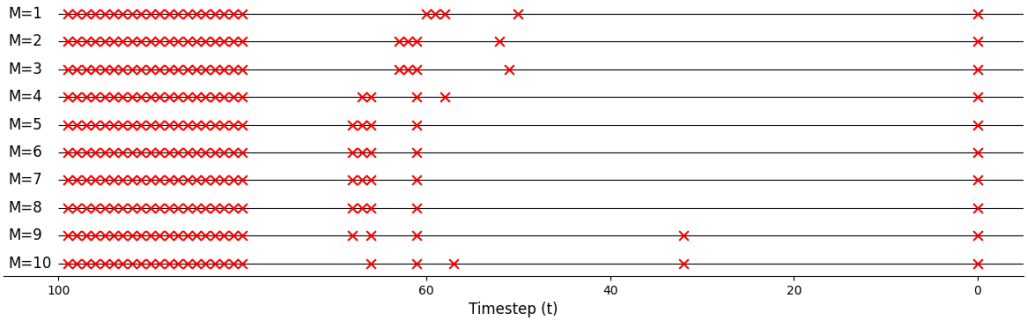

Figure 9: Optimized guidance schedules $\mathcal{G}$ for varying M (Binary MNIST, SMC-grad)

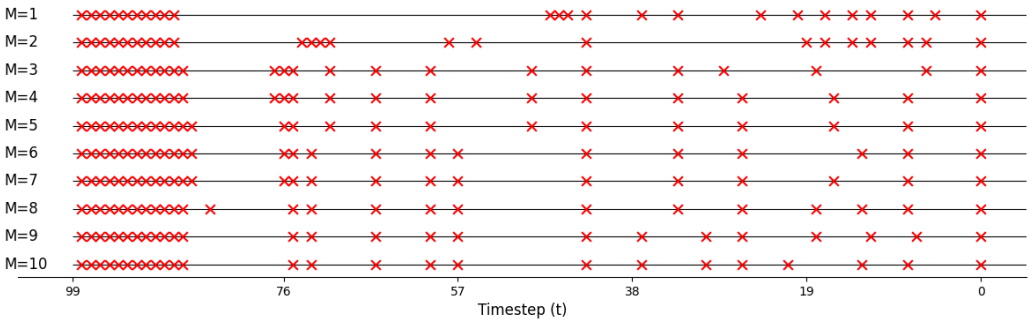

Figure 10: Optimized guidance schedules $\mathcal{G}$ for varying M (Toxic Text, SMC-grad)

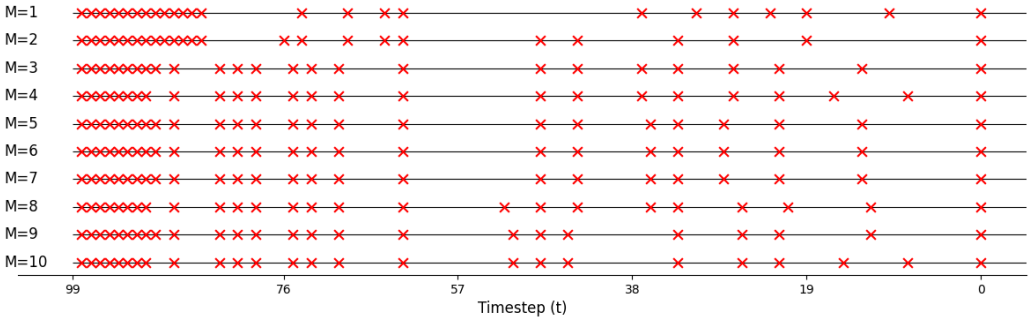

Figure 11: Optimized guidance schedules $\mathcal{G}$ for varying M (Toxic Text, SMC-amot)

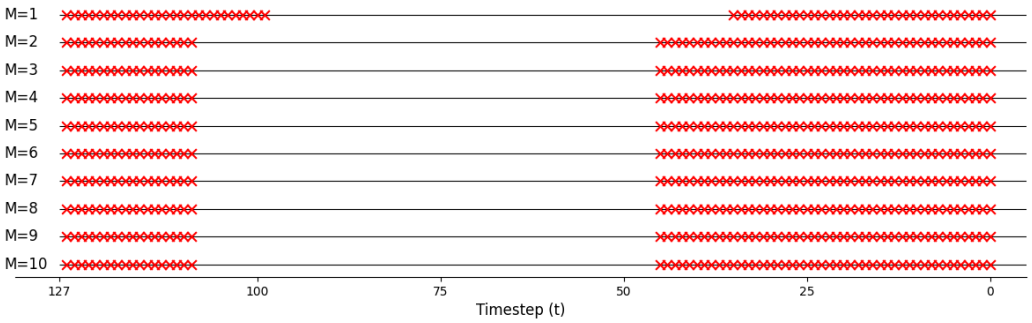

Figure 12: Optimized guidance schedules $\mathcal{G}$ for varying M (Biology Design, SMC-amot)

Table 8: Additional Binary MNIST results.

Variant	Schedule	Time [s] ↓	Unique valid ↑	Reward ↑	UTD ↓
SMC-base($T' = 10$)	Full	1.68 ± 0.05	17.15 ± 4.06	-0.0099 ± 0.0166	–
	First	1.58 ± 0.01	14.55 ± 3.47	-1.7489 ± 1.3782	7.90×10^{-1}
	Last	1.57 ± 0.01	1.80 ± 1.11	-0.6045 ± 1.5295	1.75
	Middle	1.57 ± 0.01	2.10 ± 1.55	-0.4747 ± 1.3818	9.37×10^{-1}
	Uniform	1.58 ± 0.02	8.40 ± 3.73	-0.1445 ± 0.1492	3.38×10^{-1}
	VISTA-SMC	$\mathbf{1.58 \pm 0.02}$	$\mathbf{14.45 \pm 2.91}$	$\mathbf{-0.0851 \pm 0.1641}$	1.17×10^{-1}
SMC-base($T' = 35$)	Full	1.70 ± 0.01	17.15 ± 4.06	-0.0099 ± 0.0166	–
	First	1.64 ± 0.01	17.20 ± 3.71	-0.0506 ± 0.1082	1.34×10^{-2}
	Last	1.63 ± 0.01	1.85 ± 1.09	-0.4810 ± 1.1682	1.27
	Middle	1.64 ± 0.02	2.85 ± 2.18	-0.4933 ± 1.0068	6.76×10^{-1}
	Uniform	1.63 ± 0.01	15.75 ± 3.93	-0.0245 ± 0.0433	2.06×10^{-1}
	VISTA-SMC	$\mathbf{1.64 \pm 0.01}$	$\mathbf{17.05 \pm 3.83}$	$\mathbf{-0.0386 \pm 0.0614}$	7.53×10^{-3}
SMC-grad($T' = 10$)	Full	4.90 ± 0.17	16.00 ± 3.83	-0.0041 ± 0.0077	–
	First	1.98 ± 0.03	15.80 ± 2.24	-1.3131 ± 0.8713	4.70×10^{-1}
	Last	1.95 ± 0.02	1.80 ± 1.11	-0.5745 ± 1.5352	1.75
	Middle	1.96 ± 0.02	2.10 ± 1.21	-0.4470 ± 1.6558	8.79×10^{-1}
	Uniform	2.02 ± 0.03	8.90 ± 4.14	-0.0416 ± 0.0720	2.27×10^{-1}
	VISTA-SMC	$\mathbf{1.99 \pm 0.03}$	$\mathbf{14.25 \pm 3.77}$	$\mathbf{-0.2325 \pm 0.5236}$	5.06×10^{-2}
SMC-grad($T' = 35$)	Full	5.15 ± 0.31	16.00 ± 3.83	-0.0041 ± 0.0077	–
	First	2.93 ± 0.17	16.50 ± 3.94	-0.0370 ± 0.0783	1.95×10^{-2}
	Last	2.86 ± 0.18	1.90 ± 1.07	-0.5610 ± 1.9999	1.27
	Middle	2.86 ± 0.14	2.70 ± 2.11	-0.3526 ± 1.2039	6.26×10^{-1}
	Uniform	2.80 ± 0.07	16.45 ± 3.10	-0.0054 ± 0.0095	1.19×10^{-1}
	VISTA-SMC	$\mathbf{2.93 \pm 0.15}$	$\mathbf{16.40 \pm 3.72}$	$\mathbf{-0.0230 \pm 0.0653}$	1.15×10^{-3}

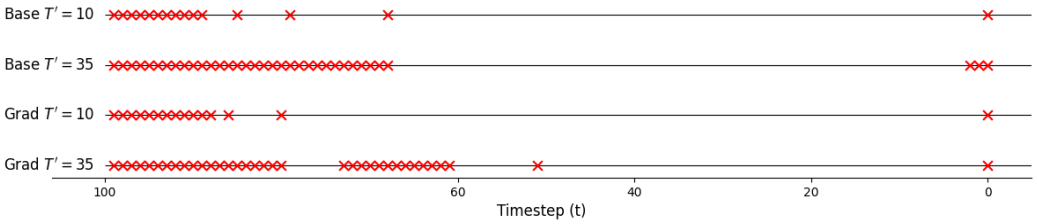

Figure 13: Picture of $\mathcal{G}$ in Tab 8

VISTA-SMC and the "First" heuristic—both of which exhibit the lowest UTD values—consistently outperform other reduced schedules.

For the Toxic Text Generation task, we follow the experimental setup in Section 5.2. Each scheduling method is evaluated over 15 independent runs, generating a total of 120 samples per variant. As shown in Table 9, VISTA-SMC and the "Uniform" heuristic, which maintain the maximum $\hat{S}(\mathcal{G})$ scores, consistently yield superior alignment quality compared to other baseline schedules. These cross-budget evaluations further confirm that the UTD metric is a robust indicator of task performance across diverse discrete diffusion domains.

C Societal Impacts

Positive impacts. VISTA-SMC reduces the cost of SMC-based reward guidance. This can make inference-time alignment more accessible to researchers with limited hardware and can reduce the energy cost of repeated reward-guided generation.

Potential negative impacts. More efficient control of generative models could be misused to generate harmful or manipulative content at lower cost. The toxic-text experiment is included only as a controlled benchmark for testing reward steering; it is not an endorsement of generating toxic content.

Table 9: Additional toxic text generation results.

Variant	Schedule	Time [s] ↓	Toxic [%] ↑	Toxic-ext [%] ↑	Dist-1/2/3 [%] ↑	PPL ↓	$\widehat{S}(\mathcal{G})$ ↑
SMC-grad ($T' = 10$)	Full	237.81 ± 1.77	100.00	90.83	51.3/79.7/85.6	117.6	–
	First	33.83 ± 0.41	72.50	70.00	63.0/93.6/95.9	123.0	3.157
	Last	33.88 ± 0.46	29.17	30.00	17.0/24.0/26.6	119.6	8.829×10^{-13}
	Middle	33.91 ± 0.33	83.33	66.67	46.6/76.9/86.0	109.4	4.664
	Uniform	33.63 ± 0.30	96.67	77.50	47.6/75.0/81.4	83.2	8.893
	VISTA-SMC	$\mathbf{33.55 \pm 0.33}$	**89.17**	**83.33**	**45.1/70.4/75.2**	**93.5**	**9.911**
SMC-grad ($T' = 35$)	Full	210.72 ± 0.24	100.00	90.83	51.3/79.7/85.6	117.6	–
	First	81.98 ± 0.34	80.83	81.67	59.8/91.4/94.8	108.9	3.188
	Last	81.92 ± 0.57	94.17	65.83	25.3/41.4/50.1	141.9	6.349×10^{-13}
	Middle	82.44 ± 0.66	86.67	80.83	47.5/78.9/87.6	147.2	4.903
	Uniform	82.10 ± 0.94	100.00	88.33	50.8/78.2/83.5	106.7	7.296
	VISTA-SMC	$\mathbf{82.20 \pm 0.96}$	**99.17**	**88.33**	**49.5/77.2/82.1**	**114.4**	**9.282**
SMC-amot ($T' = 10$)	Full	38.31 ± 1.64	100.00	87.50	27.0/43.5/49.4	58.6	–
	First	16.35 ± 0.89	13.33	26.67	59.8/90.3/93.9	73.4	8.725×10^{-13}
	Last	16.27 ± 0.88	47.50	51.67	13.7/18.6/21.0	93.5	8.829×10^{-13}
	Middle	16.09 ± 0.68	31.67	34.17	45.9/75.9/84.7	78.3	1.223×10^{1}
	Uniform	16.74 ± 1.12	79.17	93.33	25.2/40.0/45.6	63.3	2.256×10^{1}
	VISTA-SMC	$\mathbf{16.16 \pm 0.56}$	**85.83**	**80.00**	**32.2/51.9/58.4**	**92.4**	$\mathbf{2.462\times10^{1}}$
SMC-amot ($T' = 35$)	Full	38.02 ± 0.40	100.00	87.50	27.0/43.5/49.4	58.6	–
	First	22.12 ± 0.23	61.67	77.50	52.7/85.7/92.1	66.9	4.516×10^{-2}
	Last	22.34 ± 0.68	80.00	80.00	17.6/26.5/31.3	101.6	6.349×10^{-13}
	Middle	22.29 ± 0.23	70.83	76.67	43.4/73.0/82.9	79.5	1.275×10^{1}
	Uniform	22.32 ± 0.21	79.17	80.00	28.7/47.2/53.5	93.6	1.851×10^{1}
	VISTA-SMC	$\mathbf{22.42 \pm 0.45}$	**86.67**	**93.33**	**37.7/58.9/65.0**	**77.9**	$\mathbf{2.462\times10^{1}}$

Mitigation. The main risk depends on the reward model and application. We recommend using VISTA-SMC with ethically aligned reward functions, safety filters, monitoring, and application-specific release controls. The scheduler itself is neutral: it decides when a given reward should be applied, but it does not define the reward.